

Problem 282: Shepherders

Difficulty: Easy

Author: Tyler Albaugh, Denver, Colorado, United States

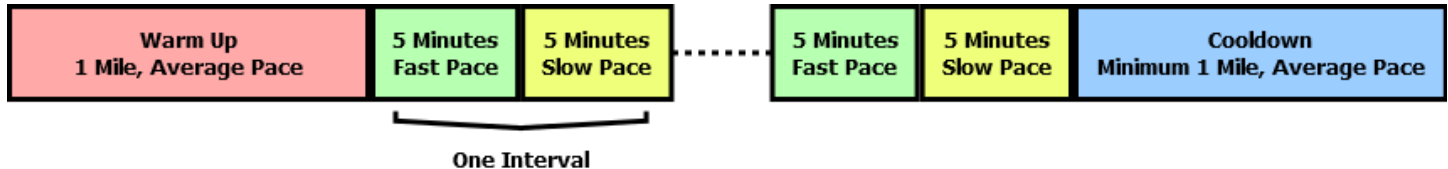
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Problem Background

Lockheed Martin's campus in Waterton, Colorado has a running club to help build community and promote an active lifestyle. Named the Shepherders, after the bighorn sheep that grace the club's favorite running trail through Waterton Canyon, the club organizes various runs to encourage employees to get moving!

Problem Description

The Shepherders want your help planning a training run. The workout should include exactly one mile of warm-up, some timed interval repeats, and a cool-down of at least one mile. The interval repeats consist of running for 5 minutes at a fast pace and then 5 minutes at a slow pace. The organizers need to know how many intervals each member of the club can run, given the trail's distance.



Each member knows their own average pace in minutes and seconds per mile. Their warm-up and cool-down are run at exactly average pace. However, for the intervals, the fast pace is meant to be one minute per mile faster than their average and the slow pace is meant to be one minute per mile slower than their average.

Given a runner's average pace and the length of the trail they're using, calculate the maximum number of intervals they can run on the trail using the workout schedule above.

Sample Input

The first line of your program's input, received from the standard input channel, will contain a positive integer representing the number of test cases. Each test case will include a single line with two values separated by a space:

- A positive decimal number representing the total length of the trail in miles
- A time, in MM:SS format, representing the runner's average pace in minutes per mile

2
9.0 7:20
7.3 5:00

Sample Output

For each test case, your program must print a single line with an integer representing the maximum number of possible intervals for each runner. This will be a minimum of 1 for each test case.

5
2